HACKATHON PRESENTATION

AI & Smart Governance Track

CivicPriority AI

Transforming raw citizen feedback into prioritized, actionable municipal operations using automated intelligence.

⚡ Real-Time Civic Triage System

THE CHALLENGE

Municipal Bottlenecks & Critical Delays

Unstructured Reports

Thousands of unorganized citizen complaints flood municipal channels daily through multiple fragmented portals.

Delayed Action

Critical infrastructure safety hazards sit trapped in backlogs behind low-priority administrative tasks.

Zero Transparency

Citizens lack real-time visibility into issue tracking, work assignment, and field resolution status.

THE CORE INNOVATION

AI-Powered Priority Matrix

Natural Language Processing

Extracts urgency, exact location, and severity automatically from unstructured text and voice intake.

Dynamic Priority Scoring

Calculates real-time threat level based on immediate public safety risk and spatial proximity density.

Automated Dispatch Routing

Directs parsed reports straight to correct municipal departments without manual human triage overhead.



Automated Triage Acceleration

Significantly cuts emergency response cycles by immediately separating routine requests from urgent public safety hazards.

TECHNICAL BLUEPRINT

End-to-End System Architecture

Citizen Portal

Multilingual voice & text report intake.

AI Pipeline

Intent classification & geo deduplication.

Triage Engine

Calculates community pressure score.

Dispatch Panel

Work order generation & field routing.

INTEROPERABILITY

Seamless Municipal Integration

Legacy CRM Connectors

Syncs bidirectionally with existing city portals and emergency 311 systems without infrastructure overhauls.

GIS Mapping APIs

Overlays reported incidents onto spatial infrastructure maps in real time to visualize high-density issue hotspots.

Automated Webhooks

Triggers instant SMS and WhatsApp notifications to citizens upon status changes and resolution steps.

ETHICAL FRAMEWORK

Responsible AI & Privacy Safeguards

PII Redaction

Automatic stripping of sensitive personally identifiable information immediately at point of intake.

Bias Mitigation

Demographic-blind priority scoring ensures equitable, fair resource allocation across all neighborhoods.

Human Verification

Critical infrastructure dispatches require final field officer confirmation prior to unit deployment.

EXECUTION STRATEGY

Scalable Deployment Roadmap

PHASE 01

Pilot Deployment

Launch in 2 target municipal districts to train local language models on real regional feedback.

PHASE 02

City-Wide Rollout

Integrate with main emergency lines and deploy automated work order dispatch modules.

PHASE 03

Predictive Maintenance

Analyze historical report patterns to proactively prevent recurring municipal infrastructure issues.

TECHNOLOGY STACK

Core Machine Learning & Engineering Framework

Python & Pandas

Core programming foundation for high-performance data processing and ingestion pipelines.

Scikit-Learn

Machine learning algorithms for supervised classification and dynamic risk priority modeling.

TF-IDF Processing

Natural language vectorization for extracting feature importance from raw unstructured text.

Explainability (SHARP / LIME)

Interpretable AI layers showing exact feature weights behind automated triage decisions.

Streamlit Dashboard

Interactive real-time web console for municipal dispatchers and operational monitoring.

Evaluation Metrics

Rigorously validated via Precision, Recall, F1-Score, and Confusion Matrix diagnostics.

NEXT STEPS

Building Smarter, Responsive Cities

Empowering communities with real-time AI prioritization and transparent public service response.

Questions & Discussion